

MindEase

The AI Learning Layer That Adapts To Every Brain

A browser extension that transforms any educational content in real time to match how you personally learn and synthesizes everything you studied into a personalized session summary at the end. (we can focus on dyslexia and ADHD diagnosed students)

The Problem

Educational content is built for one type of brain. Dense PDFs, fast lecture videos, information-heavy websites, same format, delivered the same way, to every student. Average students struggle. Students with dyslexia and ADHD fail entirely.

Current accessibility tools apply the same fix to everyone. That's not personalization. That's still ignoring the individual.

SDG 4.5 - eliminate education disparities for vulnerable groups, remains unmet because solutions are generic, not personal.

The Solution - Three Layers

Layer 1 - Real Time Content Transformation Extension intercepts any educational content:

- PDFs → chunked, restructured, simplified
- Videos → adaptive captions + paced concept breakdowns
- Websites → cognitive load reduced, distractions stripped
- Lectures → live summarization as student follows along

Transformation isn't the same for everyone - it's driven by YOUR cognitive profile.

Layer 2 - Adaptive Cognitive Profiling On first use, a 5-minute assessment identifies how you specifically process information. Not just "do you have dyslexia" but:

- do you process better with visuals or text?
- what's your attention span for continuous reading?
- do you need concept anchors before details?
- how fast do you process dense information?

Every interaction you make - highlights, re-reads, pauses, skips - feeds an RL agent that continuously refines your profile. The extension gets smarter about YOU every session.

Layer 3 - Session Memory + Synthesis At the end of every study session, Our extension generates a personalized knowledge artifact:

- key concepts you genuinely engaged with
- gaps you skimmed over, flagged for review
- connections drawn between different sources studied
- a clean study card tailored to your cognitive style

Not a generic summary. YOUR summary, built from how YOUR brain moved through the material.

Why It's Novel

Three things that don't exist together anywhere:

1. **Content-agnostic transformation** - one tool, every format
2. **RL-based personal cognitive profiling** - adapts to you specifically, continuously
3. **Session synthesis** - closes the learning loop, turns passive consumption into active knowledge

Microsoft Immersive Reader applies the same transformation to every user. Our extension applies a different transformation to every user, and that transformation evolves. That's not accessibility - that's personalized cognitive augmentation.

Why Now

- LLMs are finally cheap and fast enough for real-time content transformation
 - Browser extensions now have sufficient API access to run this pipeline
 - Post-COVID explosion of online learning created massive inaccessible content libraries
 - No tool currently combines transformation + personalization + synthesis in one layer
-

Who It's For

- **Every student** who wants to study smarter and retain more
 - **Students with dyslexia and ADHD** specifically - for whom this isn't a nice-to-have, it's the difference between staying in education or dropping out
 - **Global South students** studying in second or third languages with added cognitive load
-